

CONSOLIDATED VERSION



**Electric motor-operated hand-held tools, transportable tools and lawn and garden machinery – Safety –
Part 2-11: Particular requirements for hand-held reciprocating saws**



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**Electric motor-operated hand-held tools, transportable tools and lawn and garden machinery – Safety –
Part 2-11: Particular requirements for hand-held reciprocating saws**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

ELECTRIC MOTOR-OPERATED HAND-HELD TOOLS, TRANSPORTABLE TOOLS AND LAWN AND GARDEN MACHINERY – SAFETY –

Part 2-11: Particular requirements for hand-held reciprocating saws

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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This Consolidated version of IEC 62841-2-11 bears the edition number 1.1. It consists of the first edition (2015-11) [documents 116/246/FDIS and 116/256/RVD] and its amendment 1 (2018-01) [documents 116/353/FDIS and 116/360/RVD]. The technical content is identical to the base edition and its amendment.

In this Redline version, a vertical line in the margin shows where the technical content is modified by amendment 1. Additions are in green text, deletions are in strikethrough red text. A separate Final version with all changes accepted is available in this publication.

International Standard IEC 62841-2-11 has been prepared by IEC technical committee 116: Safety of motor-operated electric tools.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

This Part 2-11 is to be used in conjunction with the first edition of IEC 62841-1 (2014).

This Part 2-11 supplements or modifies the corresponding clauses in IEC 62841-1, so as to convert it into the IEC standard: Particular requirements for hand-held reciprocating saws.

Where a particular subclause of Part 1 is not mentioned in this Part 2-11, that subclause applies as far as relevant. Where this standard states “addition”, “modification” or “replacement”, the relevant text in Part 1 is to be adapted accordingly.

The following print types are used:

- requirements: in roman type;
- *test specifications: in italic type;*
- notes: in small roman type.

The terms defined in Clause 3 are printed in **bold typeface**.

Subclauses, notes and figures which are additional to those in Part 1 are numbered starting from 101.

A list of all parts of the IEC 62841 series, under the general title: *Electric motor-operated hand-held tools, transportable tools and lawn and garden machinery – Safety*, can be found on the IEC website.

The committee has decided that the contents of the base publication and its amendment will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

NOTE The attention of National Committees is drawn to the fact that equipment manufacturers and testing organizations may need a transitional period following publication of a new, amended or revised IEC publication in which to make products in accordance with the new requirements and to equip themselves for conducting new or revised tests.

It is the recommendation of the committee that the content of this publication be adopted for implementation nationally not earlier than 36 months from the date of publication.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

ELECTRIC MOTOR-OPERATED HAND-HELD TOOLS, TRANSPORTABLE TOOLS AND LAWN AND GARDEN MACHINERY – SAFETY –

Part 2-11: Particular requirements for hand-held reciprocating saws

1 Scope

This clause of Part 1 is applicable, except as follows:

Addition:

This part of IEC 62841 applies to **reciprocating saws** such as **jig saws** and **sabre saws**.

2 Normative references

This clause of Part 1 is applicable, except as follows:

Addition:

ISO 16893-1:2008, *Wood based panels – Particleboard – Part 1: Classifications*

3 Terms and definitions

This clause of Part 1 is applicable, except as follows:

Additional definitions:

3.101

reciprocating saw

tool intended for cutting various materials with a blade or blades acting in a reciprocating or oscillating motion

3.102

jig saw

reciprocating saw with a base plate that may allow bevel adjustments

Note 1 to entry: Typical designs of **jig saws** are shown in Figure 102.

3.103

sabre saw

reciprocating saw with a guide plate that may allow tilting movements

Note 1 to entry: Typical designs of **sabre saws** are shown in Figure 103.

4 General requirements

This clause of Part 1 is applicable.

5 General conditions for the tests

This clause of Part 1 is applicable, except as follows:

5.17 Addition:

The mass of the tool includes the dust extraction adapter, if any.

6 Radiation, toxicity and similar hazards

This clause of Part 1 is applicable.

7 Classification

This clause of Part 1 is applicable.

8 Marking and instructions

This clause of Part 1 is applicable, except as follows:

8.14.1 Addition:

For **reciprocating saws**, the additional safety instructions as specified in 8.14.1.101 shall be given. This part may be printed separately from the “General Power Tool Safety Warnings”.

8.14.1.101 Safety instructions for reciprocating saws

- a) **Hold the power tool by insulated gripping surfaces, when performing an operation where the cutting accessory may contact hidden wiring or its own cord.** *Cutting accessory contacting a "live" wire may make exposed metal parts of the power tool "live" and could give the operator an electric shock.*
- b) **Use clamps or another practical way to secure and support the workpiece to a stable platform.** *Holding the workpiece by hand or against your body leaves it unstable and may lead to loss of control.*

NOTE The above warning is not applicable for garden saws for tree or hedge services.

8.14.2 b) Addition:

- 101) Instruction on the correct use of the dust collection system, if any;
- 102) Information regarding the cutting capacity;
- 103) Instruction on how to adjust the tool for different saw blade positions, if applicable.

8.14.2 c) Addition:

- 101) Instruction on how to properly clean/clear the chip ejection opening, if any.

9 Protection against access to live parts

This clause of Part 1 is applicable.

10 Starting

This clause of Part 1 is applicable.

11 Input and current

This clause of Part 1 is applicable.

12 Heating

This clause of Part 1 is applicable.

13 Resistance to heat and fire

This clause of Part 1 is applicable.

14 Moisture resistance

This clause of Part 1 is applicable.

15 Resistance to rusting

This clause of Part 1 is applicable.

16 Overload protection of transformers and associated circuits

This clause of Part 1 is applicable.

17 Endurance

This clause of Part 1 is applicable.

18 Abnormal operation

This clause of Part 1 is applicable, except as follows:

18.8 Replacement of Table 4:

Table 4 – Required performance levels

Type and purpose of SCF	Minimum performance level (PL)
Power switch – prevent unwanted switch-on	b
Power switch – provide desired switch-off for jig saws	a
Power switch – provide desired switch-off for sabre saws	b
Any electronic control to pass the test of 18.3	Not an SCF
Any speed limiting device	Not an SCF
Prevent exceeding thermal limits as in Clause 18	a
Prevent self-resetting as required in 23.3	b

19 Mechanical hazards

This clause of Part 1 is applicable, except as follows:

19.1 Replacement of the first paragraph:

Moving and dangerous parts other than the saw blade shall be so positioned or enclosed to provide adequate protection against personal injury. Requirements for the saw blade are given in 19.101.

19.6 This subclause is not applicable.

19.101 Saw blade barrier

1) For **jig saws**

A barrier shall be provided to prevent inadvertent contact with the cutting edge of the saw blade above the plane of the base plate and from the front of the tool. The barrier shall not prevent the visual observation of the saw blade in contact with the workpiece.

Compliance is checked by inspection and by the following test.

*The **jig saw** is set for the right-angled cut. The test probe of Figure 101a) is advanced in any single plane perpendicular to the saw blade and parallel to and above the plane of the base plate as shown in Figure 101b) and Figure 101c). The longitudinal axis of the test probe shall be perpendicular to the toothed edge of the saw blade. The test probe shall be equally shared about the central plane of the saw blade. When the test probe is moved towards the saw blade, it shall not be able to touch its toothed edge.*

2) For other types of **reciprocating saws**

If a **reciprocating saw** is designed with a grip zone adjacent and behind the saw blade, a barrier shall be provided to prevent inadvertent contact with the teeth of the saw blade in any saw blade position in accordance with 8.14.2 b) 103).

The barrier shall:

- be located between the grip zone and the teeth of the saw blade;
- have a minimum height of 6 mm above the grip surface; and
- extend a minimum of 6 mm on either side of the ~~teeth~~ central plane of the saw blade, see Figure 104.

A barrier is not required if the tool is supplied with an auxiliary front handle.

Compliance is checked by inspection and by measurement.

20 Mechanical strength

This clause of Part 1 is applicable.

20.5 Addition:

Reciprocating saws are regarded as tools that are likely to cut into concealed wiring or their own cord.

21 Construction

This clause of Part 1 is applicable, except as follows:

21.18.1 *Addition:*

For **jig saws**, power switches other than momentary power switches are permitted.

21.18.1.2 This subclause is not applicable.

21.30 *Addition:*

Reciprocating saws are regarded as tools that are likely to cut into concealed wiring or their own cord.

21.35 This subclause is not applicable.

22 Internal wiring

This clause of Part 1 is applicable.

23 Components

This clause of Part 1 is applicable, except as follows:

23.3 *Replacement of the first paragraph:*

Protection devices (e.g. overload or over-temperature protection devices) or circuits that switch off the tool shall be of the non-self-resetting type unless the tool is equipped with a **momentary power switch** with no provision for being locked in the “on” position.

24 Supply connection and external flexible cords

This clause of Part 1 is applicable.

25 Terminals for external conductors

This clause of Part 1 is applicable.

26 Provision for earthing

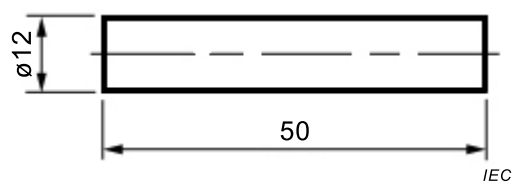
This clause of Part 1 is applicable.

27 Screws and connections

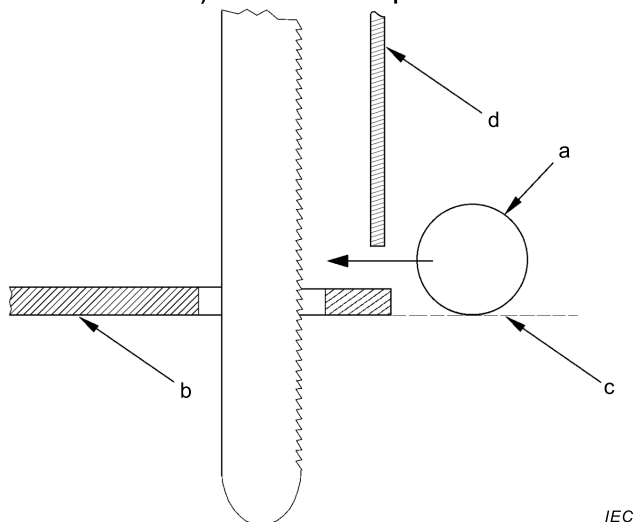
This clause of Part 1 is applicable.

28 Creepage distances, clearances and distances through insulation

This clause of Part 1 is applicable.

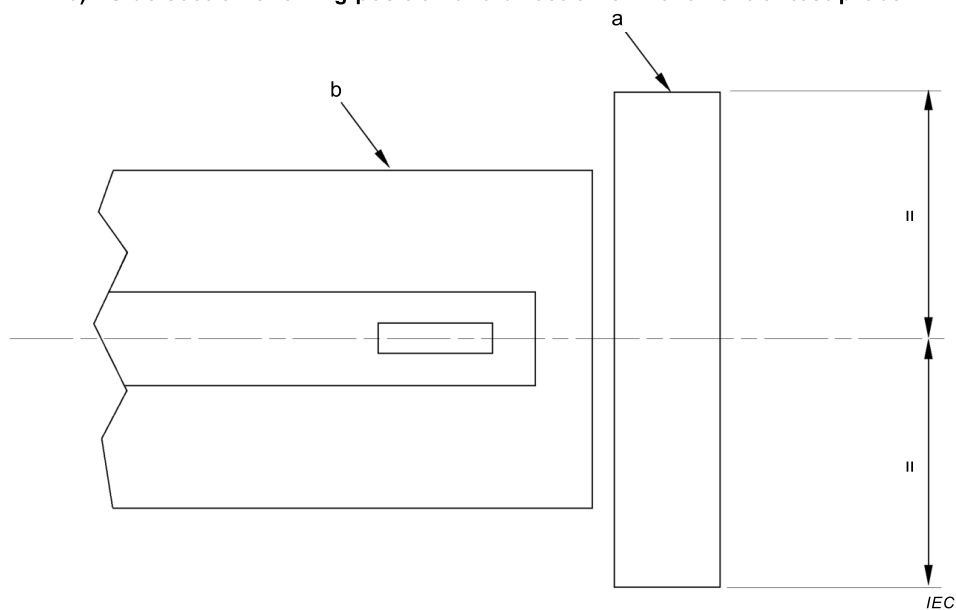


a) Details of test probe



NOTE Upper structure of saw, which should prevent the test probe touching the saw blade, is omitted for simplicity.

b) Side section showing position and direction of movement of test probe



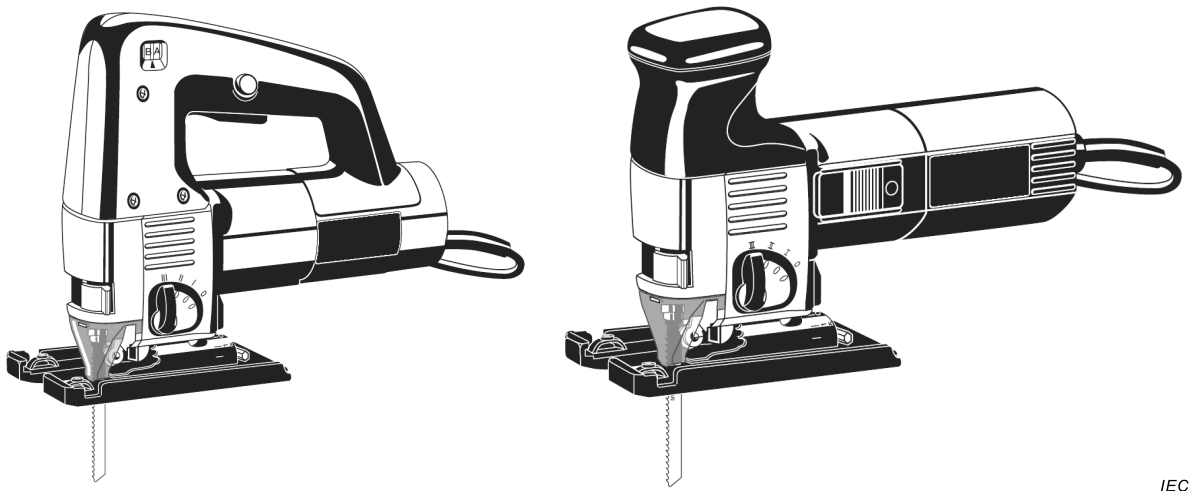
NOTE Upper structure of saw is omitted for simplicity.

c) Plan view of guide plate showing position of test probe

Key

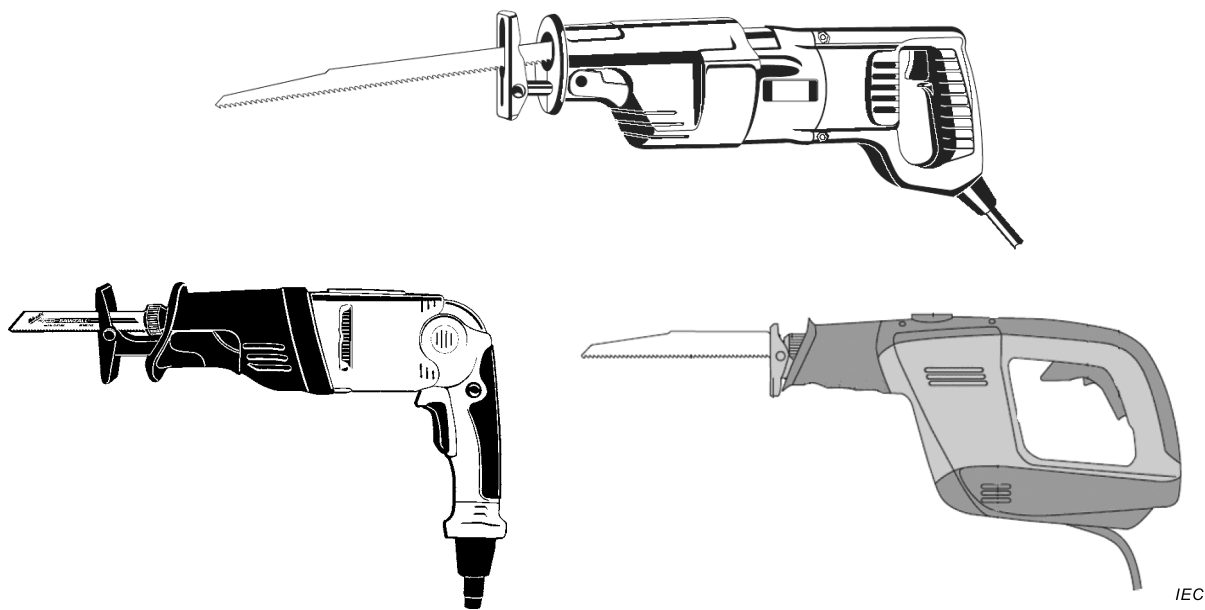
- a test probe
- b base plate
- c plane of base plate
- d barrier

Figure 101 – Test probe



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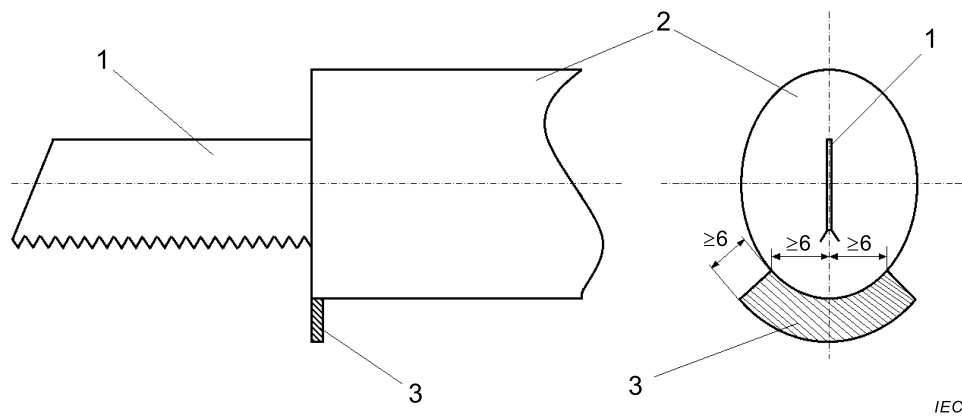
Figure 102 – Typical designs of jig saws



IEC

Figure 103 – Typical designs of sabre saws

Dimensions in millimetres



Key

- 1 saw blade
- 2 grip surface
- 3 barrier

Figure 104 – Minimum dimensions of barrier

Annexes

The annexes of Part 1 are applicable except as follows:

Annex I (informative)

Measurement of noise and vibration emissions

NOTE In Europe (EN 62841-2-11), Annex I is normative.

I.2 Noise test code (grade 2)

This clause of Part 1 is applicable except as follows:

I.2.4 Installation and mounting conditions of the power tools during noise tests

Addition:

Reciprocating saws are suspended in such a way as to correspond to **normal use**.

I.2.5 Operating conditions

Addition:

Reciprocating saws are tested at no-load, fitted with the smallest saw blade as specified for sawing chipboard. Pendulum systems, if any, are set to the maximum. Speed settings, if any, are adjusted to the highest value.

NOTE Experimental investigations have shown that the noise emission values of **reciprocating saws** at no-load and under load are very similar. For reasons of simplification, the noise emission is therefore measured at no-load.

The temperature requirements of 5.6 are not applicable.

I.2.9 Declaration and verification of noise emission values

Replacement of the second paragraph:

For a standard deviation of reproducibility of the method σ_{R0} of 1,5 dB and for a typical standard deviation of production, the values for the uncertainties, K_{pA} and K_{WA} respectively, are expected to be 5 dB.

NOTE 101 The values K_{pA} and K_{WA} are higher, because they include the noise emission under load.

I.3 Vibration

This clause of Part 1 is applicable except as follows:

I.3.3.2 Location of measurement

Addition:

Figures I.103 and I.104 show the positions for different saws.

I.3.5.3 Operating conditions

Addition:

Reciprocating saws are tested under load according to the conditions shown in Tables I.101, I.102 and I.103.

Jig saws are tested sawing both board and sheet metal. **Sabre saws** are tested cutting board and, for saws with a cutting capacity greater than or equal 100 mm in accordance with 8.14.2 b) 4), also cutting wooden beams.

Sabre saws and **jig saws** with speed setting devices shall be adjusted, as specified by the manufacturer, to the settings to cut the work piece material required in the test. If there is no specification by the manufacturer, the speed is set to maximum.

Table I.101 – Test conditions for sabre and jig saws cutting board

Orientation	For jig saws: Cutting a horizontal piece of particleboard (chipboard), class P-LB-Reg in accordance with ISO 16893-1:2008, density $(610 \pm 60) \text{ kg/m}^3$, $(38 \pm 2) \text{ mm}$ thick with a minimum length of 500 mm and a width of approximately 600 mm. The board shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.101. For sabre saws: Cutting a vertical piece of particleboard (chipboard), class P-LB-Reg in accordance with ISO 16893-1:2008, density $(610 \pm 60) \text{ kg/m}^3$, $(38 \pm 2) \text{ mm}$ thick with a minimum length of 500 mm and a width of approximately 600 mm. The board shall be supported on resilient material and fixed vertically by screws, clamps, air cylinders or the like to a test rig, see Figure I.102. In all cases, the board excess end shall be approximately 250 mm from the clamp and shall be readjusted at the beginning of each series of tests, which consists of five test cycles. During the entire cut, the saw blade shall be perpendicular to the board.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing chipboard. The saw blade used shall be recorded. Pendulum systems, if any, being set as specified by the manufacturer for cutting chipboard. If not specified by the manufacturer, pendulum systems are set to maximum. For sabre saws, the guide plate shall be fitted.
Feed force	For jig saws: The horizontal feed force (force in direction of the cut) applied to the tool shall be $(35 \pm 5) \text{ N}$. Excessive gripping force shall be avoided. The downward force applied to the tool shall not exceed the force to ensure contact of the base plate with the workpiece by more than 30 N. The feed force and the downward force shall be determined e.g. by means of a scale and shall be recorded. For sabre saws: The vertical feed force (downward force in direction of the cut) applied to the tool in addition to its weight shall be $(40 \pm 5) \text{ N}$. Excessive gripping force shall be avoided. The horizontal force applied to the tool shall not exceed the force to ensure contact of the guide plate with the workpiece by more than 30 N. The feed force and the horizontal force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 30 mm wide strip across the 600 mm width of the chipboard. Measurement starts when the saw blade enters the chipboard and stops when the saw blade leaves the chipboard.

Table I.102 – Test conditions for jig saws cutting sheet metal

Orientation	Cutting a horizontal piece of sheet mild steel with the minimum length of 300 mm, a minimum width of 100 mm and a thickness of approximately 3 mm. The workpiece shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.101. The metal sheet excess shall be approximately 80 mm from the clamped area and shall be readjusted at the beginning of each series of tests, which consists of five test cycles.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing mild steel. The saw blade used shall be recorded. The pendulum system, if any, shall be in the “off” position.
Feed force	The horizontal feed force (force in direction of the cut) applied to the tool shall be (35 ± 5) N. Excessive gripping force shall be avoided. The downward force applied to the tool shall not exceed the force to ensure contact of the base plate with the workpiece by more than 30 N. The feed force and the downward force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 8 mm wide strip across the 100 mm width of the metal sheet. Measurement starts when the saw blade enters the metal sheet and stops when the saw blade leaves the metal sheet.

Table I.103 – Test conditions for sabre saws cutting wooden beams

Orientation	Cutting a horizontal beam of construction wood such as fir with a cross-section of (100 ± 5) mm $\times$ (100 ± 5) mm and a minimum length of 500 mm. The beam shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.102. In all cases, the beam excess end shall be approximately 250 mm from the clamp and shall be readjusted at the beginning of each series of tests, which consists of five test cycles. During the test, the operator may use swivel action to allow proper chip ejection.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing large wooden beams. The saw blade used shall be recorded. Pendulum systems, if any, being set as specified by the manufacturer for cutting wooden beams. If not specified by the manufacturer, pendulum systems are set to maximum. The guide plate shall be fitted.
Feed force	The vertical feed force (force in direction of the cut) applied to the tool in addition to its weight shall be (40 ± 5) N. Excessive gripping force shall be avoided. The horizontal force applied to the tool shall not exceed the force to ensure contact of the guide plate with the workpiece by more than 30 N. The feed force and the horizontal force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 30 mm thick slice. Measurement starts when the saw blade enters the wood and stops when the saw blade leaves the wood.

I.3.6.1 Reported vibration value

Replacement:

Three series of five consecutive tests shall be carried out using a different operator for each series. If it can be shown that the vibration is not affected by operator characteristics, it is acceptable to perform all 15 measurements with one operator only.

The measurements are made in three axes and the results of each direction shall be combined using formula (I.3) to obtain the vibration total value a_{HV} .

The measurement result a_h shall be determined as the arithmetic mean of vibration total values over the tests and operators.

For **jig saws**, the results a_h for two operation modes shall be reported:

- $a_{h,B}$ = mean vibration “cutting board” in accordance with Table I.101.
- $a_{h,M}$ = mean vibration “cutting sheet metal” in accordance Table I.102.

For **sabre saws**, the results a_h for two operation modes shall be reported:

- $a_{h,B}$ = mean vibration “cutting board” in accordance with Table I.101.
- $a_{h,WB}$ = mean vibration “cutting wooden beam” in accordance Table I.103.

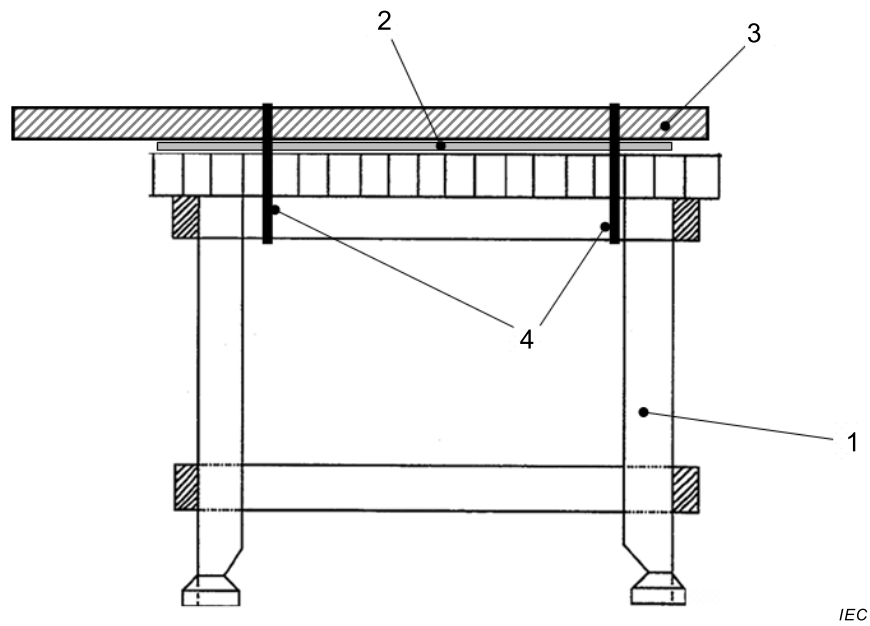
I.3.6.2 Declaration of the vibration total value

Addition:

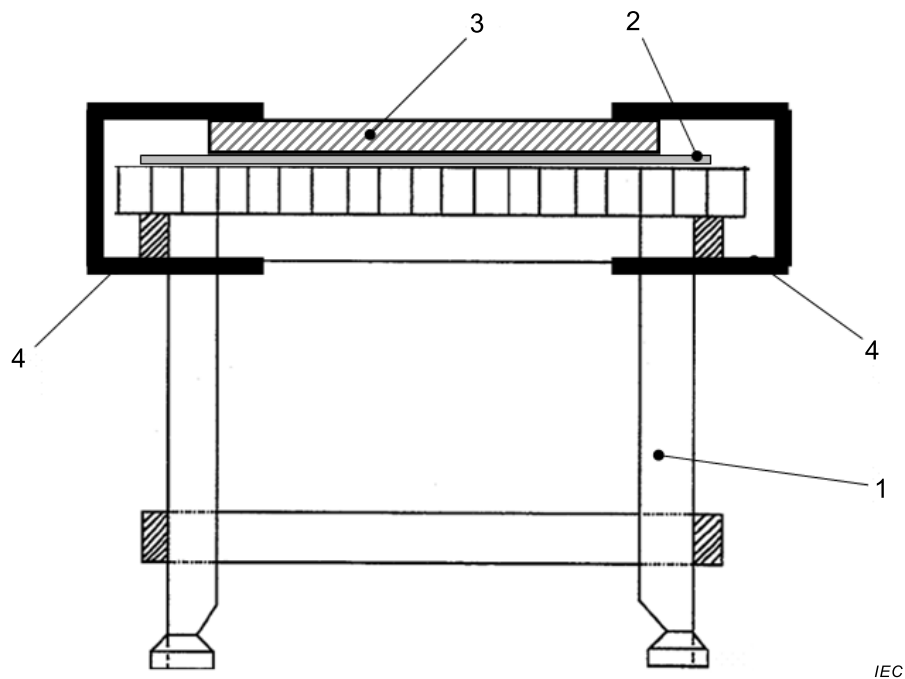
The vibration total value of the handle with the highest emission and the uncertainty K shall be declared together with the saw blade used:

- for **jig saws**
the value of $a_{h,B}$, with the work mode description “cutting boards with saw blade ...” and the value of $a_{h,M}$, with the work mode description “cutting sheet metal with saw blade ...”;
- for **sabre saws**
the value of $a_{h,B}$, with the work mode description “cutting boards with saw blade ...” and the value of $a_{h,WB}$, with the work mode description “cutting wooden beams with saw blade ...”.

NOTE The saw blade significantly influences the vibration behaviour. Therefore the combination of the vibration value along with the saw blade used for the test provides important information.



a) Front view

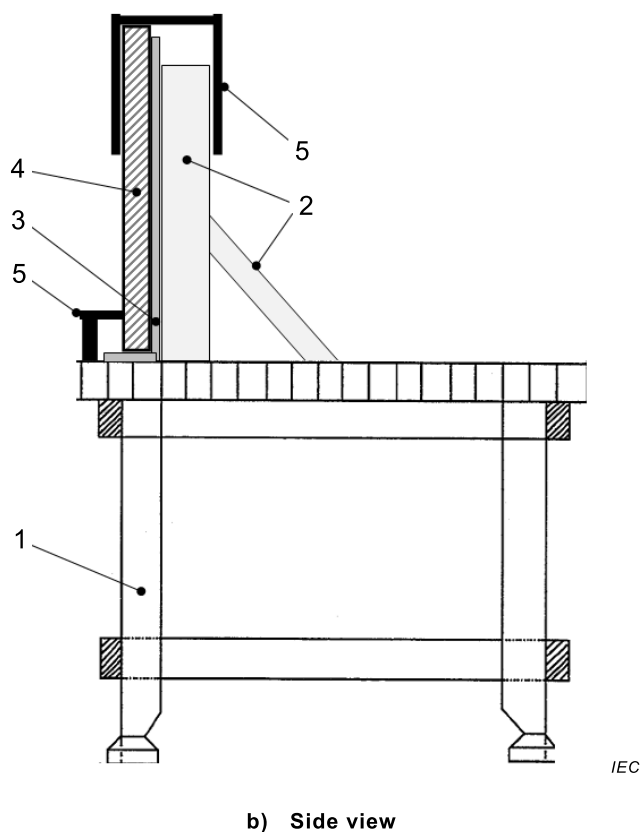
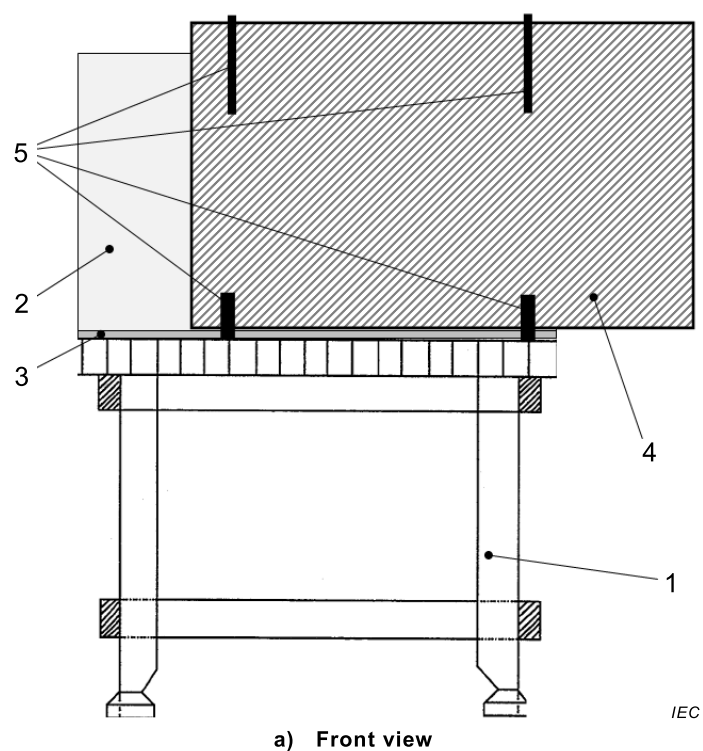


b) Side view

Key

- 1 test bench of Figure I.1
- 2 resilient material
- 3 workpiece
- 4 fixing means (e.g. clamps)

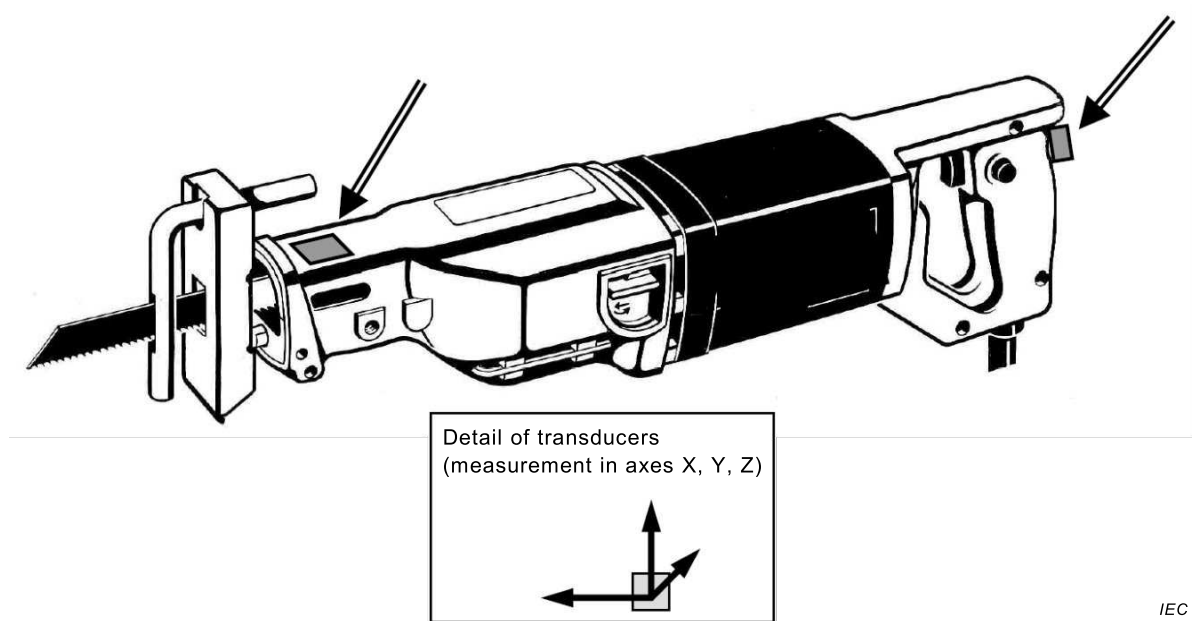
Figure I.101 – Test set-up for jig saws



Key

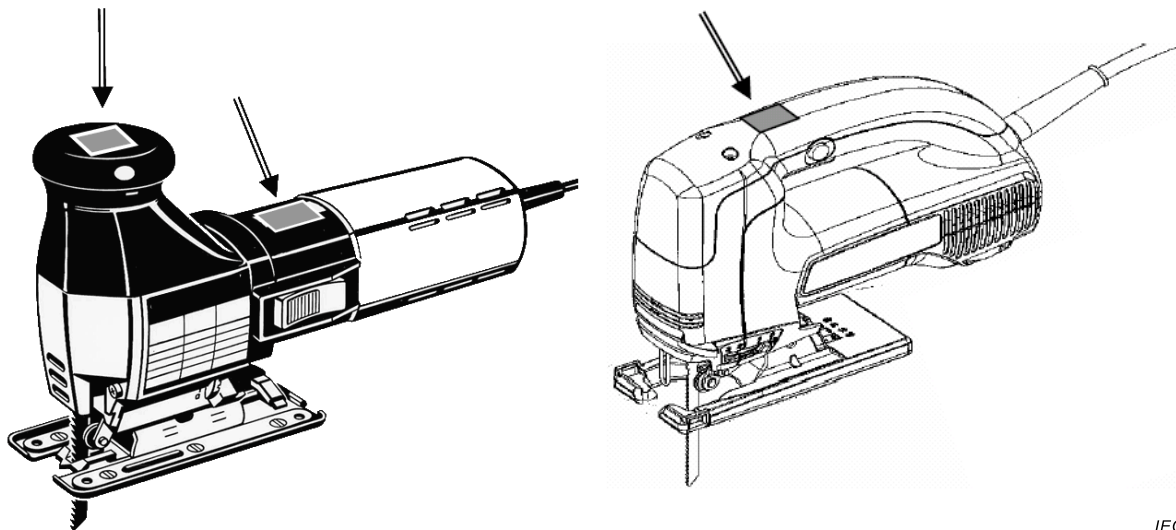
- 1 test bench of Figure I.1
- 2 vertical support, rigidly mounted on test bench
- 3 resilient material
- 4 workpiece
- 5 fixing means (e.g. clamps)

Figure I.102 – Test set-up for sabre saws



IEC

Figure I.103 – Positions of transducers for sabre saws



IEC

Figure I.104 – Positions of transducers for jig saws

Annex K (normative)

Battery tools and battery packs

K.1 Scope

Addition:

All clauses of this Part 2-11 apply unless otherwise specified in this annex.

K.8.14.1.101 *Replacement of item a):*

- 4a) **Hold the power tool by insulated gripping surfaces, when performing an operation where the cutting accessory may contact hidden wiring. Cutting accessory contacting a "live" wire may make exposed metal parts of the power tool "live" and could give the operator an electric shock.**

NOTE In Europe (EN 62841-2-11), the following additional subclause applies:

K.21.18.Z101 Isolation device

Reciprocating saws with an **integral battery** shall be equipped with an isolation device to prevent the risk of injury from mechanical hazards during **user maintenance** that

- ensures a reliable disconnection of at least one pole of the **battery** from the relevant load(s);
- has a reliable mechanical link between the manual control and the disconnecting element(s);
- is equipped with an unambiguous indication of the state of the disconnection device which corresponds to each position of its manual control (actuator);
- protects against accidental reconnection.

NOTE 101 Examples of methods to achieve this disconnection include removable jumpers, removable **batteries** or a electromechanical **power switch** with a direct mechanical link between the actuator and the contact.

NOTE 102 The risk of accidental reconnection for a **power switch** is addressed by the requirement of 21.18.1.2. The other examples in NOTE 1 achieve this by the necessary actions for reconnection.

Compliance is checked by inspection.

Annex L (normative)

Battery tools and battery packs provided with mains connection or non-isolated sources

L.1 Scope

Addition:

All clauses of this Part 2-11 apply unless otherwise specified in this annex.

NOTE In Europe (EN 62841-2-11), the following additional subclause applies:

L.21.18.Z101 Isolation device

Reciprocating saws with an **integral battery** shall be equipped with a isolation device to prevent the risk of injury from mechanical hazards during **user maintenance** that

- ensures a reliable disconnection of at least one pole of the **battery** from the relevant load(s);
- has a reliable mechanical link between the manual control and the disconnecting element(s);
- is equipped with an unambiguous indication of the state of the disconnection device which corresponds to each position of its manual control (actuator);
- protects against accidental reconnection.

NOTE 101 Examples of methods to achieve this disconnection include removable jumpers, removable **batteries** or a electromechanical **power switch** with a direct mechanical link between the actuator and the contact.

NOTE 102 The risk of accidental reconnection for a **power switch** is addressed by the requirement of 21.18.1.2. The other examples in NOTE 1 achieve this by the necessary actions for reconnection.

Compliance is checked by inspection.

Bibliography

The bibliography of Part 1 is applicable.

FINAL VERSION

**Electric motor-operated hand-held tools, transportable tools and lawn and garden machinery – Safety –
Part 2-11: Particular requirements for hand-held reciprocating saws**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

ELECTRIC MOTOR-OPERATED HAND-HELD TOOLS, TRANSPORTABLE TOOLS AND LAWN AND GARDEN MACHINERY – SAFETY –

Part 2-11: Particular requirements for hand-held reciprocating saws

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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This Consolidated version is not an official IEC Standard and has been prepared for user convenience. Only the current versions of the standard and its amendment(s) are to be considered the official documents.

This Consolidated version of IEC 62841-2-11 bears the edition number 1.1. It consists of the first edition (2015-11) [documents 116/246/FDIS and 116/256/RVD] and its amendment 1 (2018-01) [documents 116/353/FDIS and 116/360/RVD]. The technical content is identical to the base edition and its amendment.

This Final version does not show where the technical content is modified by amendment 1. A separate Redline version with all changes highlighted is available in this publication.

International Standard IEC 62841-2-11 has been prepared by IEC technical committee 116: Safety of motor-operated electric tools.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

This Part 2-11 is to be used in conjunction with the first edition of IEC 62841-1 (2014).

This Part 2-11 supplements or modifies the corresponding clauses in IEC 62841-1, so as to convert it into the IEC standard: Particular requirements for hand-held reciprocating saws.

Where a particular subclause of Part 1 is not mentioned in this Part 2-11, that subclause applies as far as relevant. Where this standard states “addition”, “modification” or “replacement”, the relevant text in Part 1 is to be adapted accordingly.

The following print types are used:

- requirements: in roman type;
- *test specifications: in italic type;*
- notes: in small roman type.

The terms defined in Clause 3 are printed in **bold typeface**.

Subclauses, notes and figures which are additional to those in Part 1 are numbered starting from 101.

A list of all parts of the IEC 62841 series, under the general title: *Electric motor-operated hand-held tools, transportable tools and lawn and garden machinery – Safety*, can be found on the IEC website.

The committee has decided that the contents of the base publication and its amendment will remain unchanged until the stability date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

NOTE The attention of National Committees is drawn to the fact that equipment manufacturers and testing organizations may need a transitional period following publication of a new, amended or revised IEC publication in which to make products in accordance with the new requirements and to equip themselves for conducting new or revised tests.

It is the recommendation of the committee that the content of this publication be adopted for implementation nationally not earlier than 36 months from the date of publication.

ELECTRIC MOTOR-OPERATED HAND-HELD TOOLS, TRANSPORTABLE TOOLS AND LAWN AND GARDEN MACHINERY – SAFETY –

Part 2-11: Particular requirements for hand-held reciprocating saws

1 Scope

This clause of Part 1 is applicable, except as follows:

Addition:

This part of IEC 62841 applies to **reciprocating saws** such as **jig saws** and **sabre saws**.

2 Normative references

This clause of Part 1 is applicable, except as follows:

Addition:

ISO 16893-1:2008, *Wood based panels – Particleboard – Part 1: Classifications*

3 Terms and definitions

This clause of Part 1 is applicable, except as follows:

Additional definitions:

3.101

reciprocating saw

tool intended for cutting various materials with a blade or blades acting in a reciprocating or oscillating motion

3.102

jig saw

reciprocating saw with a base plate that may allow bevel adjustments

Note 1 to entry: Typical designs of **jig saws** are shown in Figure 102.

3.103

sabre saw

reciprocating saw with a guide plate that may allow tilting movements

Note 1 to entry: Typical designs of **sabre saws** are shown in Figure 103.

4 General requirements

This clause of Part 1 is applicable.

5 General conditions for the tests

This clause of Part 1 is applicable, except as follows:

5.17 Addition:

The mass of the tool includes the dust extraction adapter, if any.

6 Radiation, toxicity and similar hazards

This clause of Part 1 is applicable.

7 Classification

This clause of Part 1 is applicable.

8 Marking and instructions

This clause of Part 1 is applicable, except as follows:

8.14.1 Addition:

For **reciprocating saws**, the additional safety instructions as specified in 8.14.1.101 shall be given. This part may be printed separately from the “General Power Tool Safety Warnings”.

8.14.1.101 Safety instructions for reciprocating saws

- a) **Hold the power tool by insulated gripping surfaces, when performing an operation where the cutting accessory may contact hidden wiring or its own cord.** *Cutting accessory contacting a "live" wire may make exposed metal parts of the power tool "live" and could give the operator an electric shock.*
- b) **Use clamps or another practical way to secure and support the workpiece to a stable platform.** *Holding the workpiece by hand or against your body leaves it unstable and may lead to loss of control.*

NOTE The above warning is not applicable for garden saws for tree or hedge services.

8.14.2 b) Addition:

- 101) Instruction on the correct use of the dust collection system, if any;
- 102) Information regarding the cutting capacity;
- 103) Instruction on how to adjust the tool for different saw blade positions, if applicable.

8.14.2 c) Addition:

- 101) Instruction on how to properly clean/clear the chip ejection opening, if any.

9 Protection against access to live parts

This clause of Part 1 is applicable.

10 Starting

This clause of Part 1 is applicable.

11 Input and current

This clause of Part 1 is applicable.

12 Heating

This clause of Part 1 is applicable.

13 Resistance to heat and fire

This clause of Part 1 is applicable.

14 Moisture resistance

This clause of Part 1 is applicable.

15 Resistance to rusting

This clause of Part 1 is applicable.

16 Overload protection of transformers and associated circuits

This clause of Part 1 is applicable.

17 Endurance

This clause of Part 1 is applicable.

18 Abnormal operation

This clause of Part 1 is applicable, except as follows:

18.8 Replacement of Table 4:

Table 4 – Required performance levels

Type and purpose of SCF	Minimum performance level (PL)
Power switch – prevent unwanted switch-on	b
Power switch – provide desired switch-off for jig saws	a
Power switch – provide desired switch-off for sabre saws	b
Any electronic control to pass the test of 18.3	Not an SCF
Any speed limiting device	Not an SCF
Prevent exceeding thermal limits as in Clause 18	a
Prevent self-resetting as required in 23.3	b

19 Mechanical hazards

This clause of Part 1 is applicable, except as follows:

19.1 *Replacement of the first paragraph:*

Moving and dangerous parts other than the saw blade shall be so positioned or enclosed to provide adequate protection against personal injury. Requirements for the saw blade are given in 19.101.

19.6 This subclause is not applicable.

19.101 **Saw blade barrier**

1) For **jig saws**

A barrier shall be provided to prevent inadvertent contact with the cutting edge of the saw blade above the plane of the base plate and from the front of the tool. The barrier shall not prevent the visual observation of the saw blade in contact with the workpiece.

Compliance is checked by inspection and by the following test.

*The **jig saw** is set for the right-angled cut. The test probe of Figure 101a) is advanced in any single plane perpendicular to the saw blade and parallel to and above the plane of the base plate as shown in Figure 101b) and Figure 101c). The longitudinal axis of the test probe shall be perpendicular to the toothed edge of the saw blade. The test probe shall be equally shared about the central plane of the saw blade. When the test probe is moved towards the saw blade, it shall not be able to touch its toothed edge.*

2) For other types of **reciprocating saws**

If a **reciprocating saw** is designed with a grip zone adjacent and behind the saw blade, a barrier shall be provided to prevent inadvertent contact with the teeth of the saw blade in any saw blade position in accordance with 8.14.2 b) 103).

The barrier shall:

- be located between the grip zone and the teeth of the saw blade;
- have a minimum height of 6 mm above the grip surface; and
- extend a minimum of 6 mm on either side of the central plane of the saw blade, see Figure 104.

A barrier is not required if the tool is supplied with an auxiliary front handle.

Compliance is checked by inspection and by measurement.

20 Mechanical strength

This clause of Part 1 is applicable.

20.5 *Addition:*

Reciprocating saws are regarded as tools that are likely to cut into concealed wiring or their own cord.

21 Construction

This clause of Part 1 is applicable, except as follows:

21.18.1 *Addition:*

For **jig saws**, power switches other than momentary power switches are permitted.

21.18.1.2 This subclause is not applicable.

21.30 *Addition:*

Reciprocating saws are regarded as tools that are likely to cut into concealed wiring or their own cord.

21.35 This subclause is not applicable.

22 Internal wiring

This clause of Part 1 is applicable.

23 Components

This clause of Part 1 is applicable, except as follows:

23.3 *Replacement of the first paragraph:*

Protection devices (e.g. overload or over-temperature protection devices) or circuits that switch off the tool shall be of the non-self-resetting type unless the tool is equipped with a **momentary power switch** with no provision for being locked in the “on” position.

24 Supply connection and external flexible cords

This clause of Part 1 is applicable.

25 Terminals for external conductors

This clause of Part 1 is applicable.

26 Provision for earthing

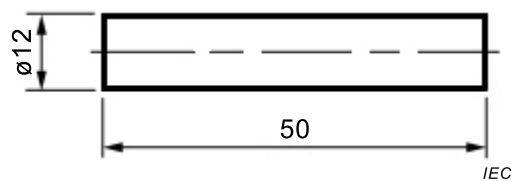
This clause of Part 1 is applicable.

27 Screws and connections

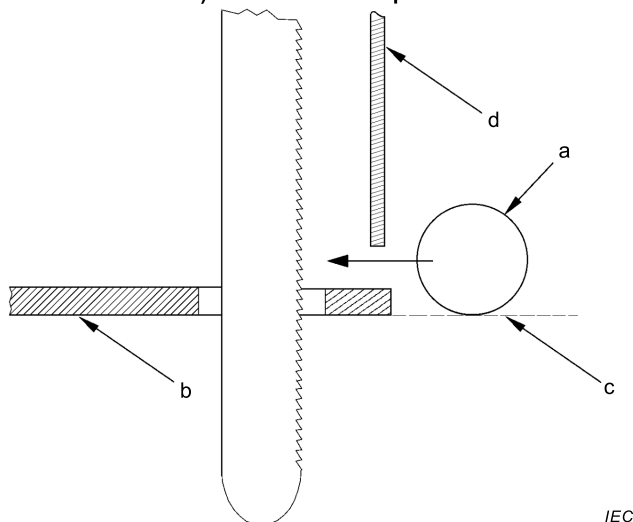
This clause of Part 1 is applicable.

28 Creepage distances, clearances and distances through insulation

This clause of Part 1 is applicable.

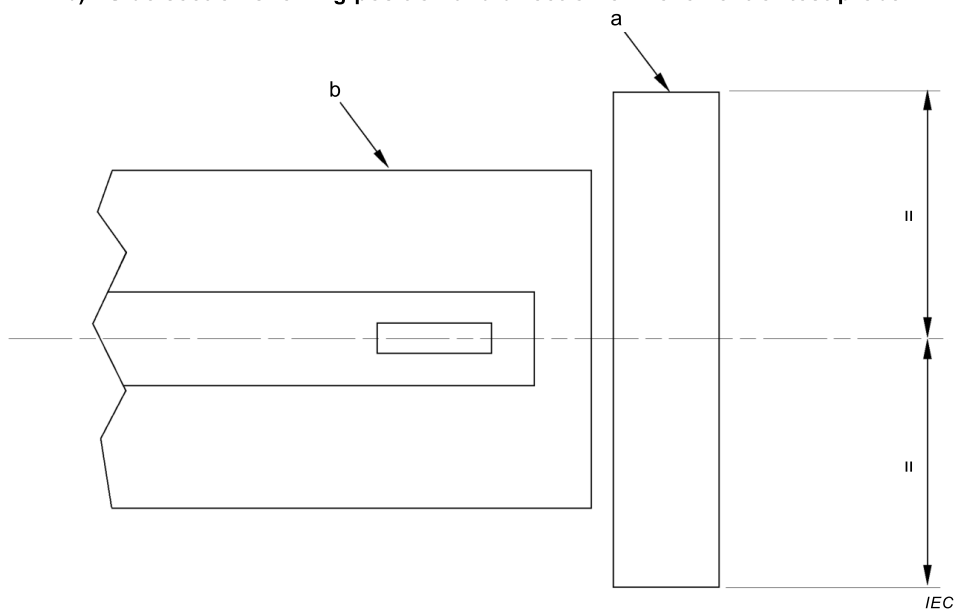


a) Details of test probe



NOTE Upper structure of saw, which should prevent the test probe touching the saw blade, is omitted for simplicity.

b) Side section showing position and direction of movement of test probe



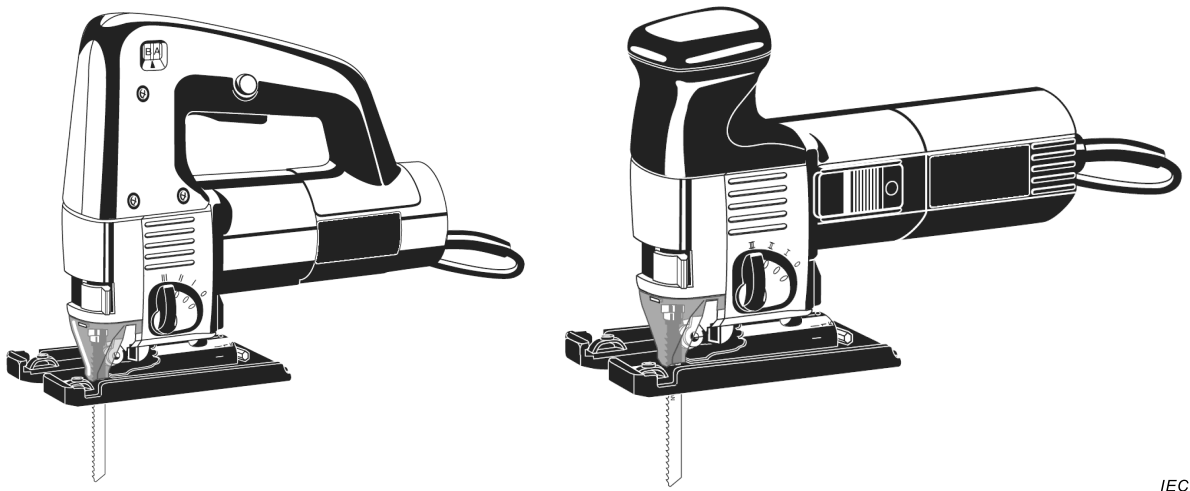
NOTE Upper structure of saw is omitted for simplicity.

c) Plan view of guide plate showing position of test probe

Key

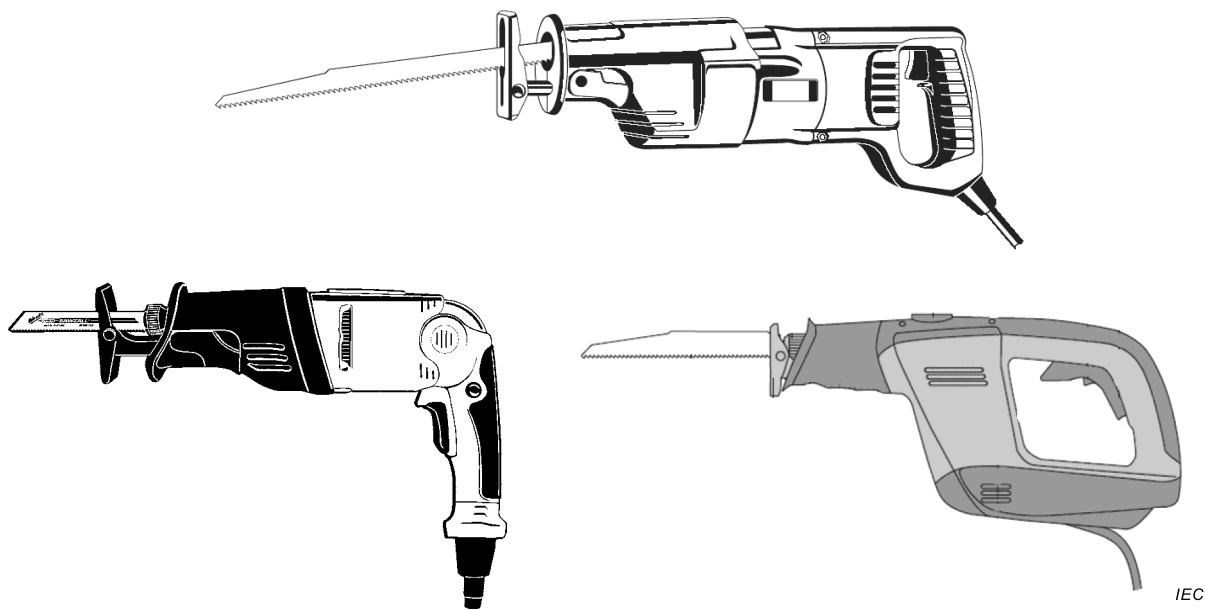
- a test probe
- b base plate
- c plane of base plate
- d barrier

Figure 101 – Test probe



IEC

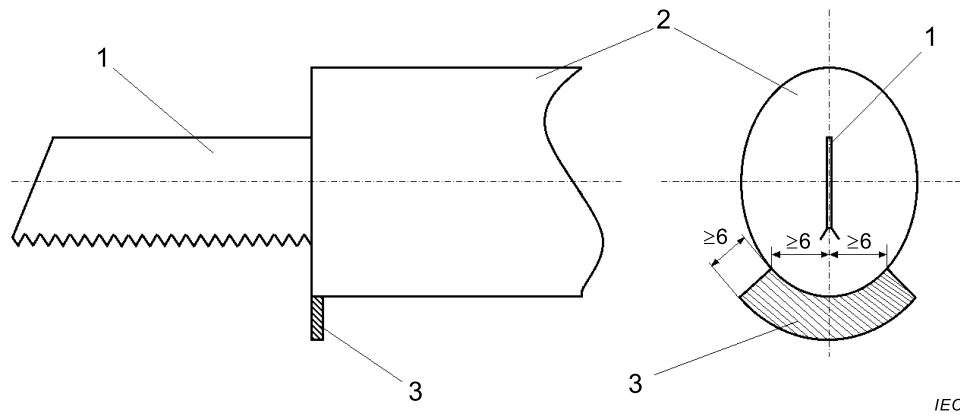
Figure 102 – Typical designs of jig saws



IEC

Figure 103 – Typical designs of sabre saws

Dimensions in millimetres



Key

- 1 saw blade
- 2 grip surface
- 3 barrier

Figure 104 – Minimum dimensions of barrier

Annexes

The annexes of Part 1 are applicable except as follows:

Annex I (informative)

Measurement of noise and vibration emissions

NOTE In Europe (EN 62841-2-11), Annex I is normative.

I.2 Noise test code (grade 2)

This clause of Part 1 is applicable except as follows:

I.2.4 Installation and mounting conditions of the power tools during noise tests

Addition:

Reciprocating saws are suspended in such a way as to correspond to **normal use**.

I.2.5 Operating conditions

Addition:

Reciprocating saws are tested at no-load, fitted with the smallest saw blade as specified for sawing chipboard. Pendulum systems, if any, are set to the maximum. Speed settings, if any, are adjusted to the highest value.

NOTE Experimental investigations have shown that the noise emission values of **reciprocating saws** at no-load and under load are very similar. For reasons of simplification, the noise emission is therefore measured at no-load.

The temperature requirements of 5.6 are not applicable.

I.2.9 Declaration and verification of noise emission values

Replacement of the second paragraph:

For a standard deviation of reproducibility of the method σ_{R0} of 1,5 dB and for a typical standard deviation of production, the values for the uncertainties, K_{pA} and K_{WA} respectively, are expected to be 5 dB.

NOTE 101 The values K_{pA} and K_{WA} are higher, because they include the noise emission under load.

I.3 Vibration

This clause of Part 1 is applicable except as follows:

I.3.3.2 Location of measurement

Addition:

Figures I.103 and I.104 show the positions for different saws.

I.3.5.3 Operating conditions

Addition:

Reciprocating saws are tested under load according to the conditions shown in Tables I.101, I.102 and I.103.

Jig saws are tested sawing both board and sheet metal. **Sabre saws** are tested cutting board and, for saws with a cutting capacity greater than or equal 100 mm in accordance with 8.14.2 b) 4), also cutting wooden beams.

Sabre saws and **jig saws** with speed setting devices shall be adjusted, as specified by the manufacturer, to the settings to cut the work piece material required in the test. If there is no specification by the manufacturer, the speed is set to maximum.

Table I.101 – Test conditions for sabre and jig saws cutting board

Orientation	For jig saws: Cutting a horizontal piece of particleboard (chipboard), class P-LB-Reg in accordance with ISO 16893-1:2008, density $(610 \pm 60) \text{ kg/m}^3$, $(38 \pm 2) \text{ mm}$ thick with a minimum length of 500 mm and a width of approximately 600 mm. The board shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.101. For sabre saws: Cutting a vertical piece of particleboard (chipboard), class P-LB-Reg in accordance with ISO 16893-1:2008, density $(610 \pm 60) \text{ kg/m}^3$, $(38 \pm 2) \text{ mm}$ thick with a minimum length of 500 mm and a width of approximately 600 mm. The board shall be supported on resilient material and fixed vertically by screws, clamps, air cylinders or the like to a test rig, see Figure I.102. In all cases, the board excess end shall be approximately 250 mm from the clamp and shall be readjusted at the beginning of each series of tests, which consists of five test cycles. During the entire cut, the saw blade shall be perpendicular to the board.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing chipboard. The saw blade used shall be recorded. Pendulum systems, if any, being set as specified by the manufacturer for cutting chipboard. If not specified by the manufacturer, pendulum systems are set to maximum. For sabre saws, the guide plate shall be fitted.
Feed force	For jig saws: The horizontal feed force (force in direction of the cut) applied to the tool shall be $(35 \pm 5) \text{ N}$. Excessive gripping force shall be avoided. The downward force applied to the tool shall not exceed the force to ensure contact of the base plate with the workpiece by more than 30 N. The feed force and the downward force shall be determined e.g. by means of a scale and shall be recorded. For sabre saws: The vertical feed force (downward force in direction of the cut) applied to the tool in addition to its weight shall be $(40 \pm 5) \text{ N}$. Excessive gripping force shall be avoided. The horizontal force applied to the tool shall not exceed the force to ensure contact of the guide plate with the workpiece by more than 30 N. The feed force and the horizontal force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 30 mm wide strip across the 600 mm width of the chipboard. Measurement starts when the saw blade enters the chipboard and stops when the saw blade leaves the chipboard.

Table I.102 – Test conditions for jig saws cutting sheet metal

Orientation	Cutting a horizontal piece of sheet mild steel with the minimum length of 300 mm, a minimum width of 100 mm and a thickness of approximately 3 mm. The workpiece shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.101. The metal sheet excess shall be approximately 80 mm from the clamped area and shall be readjusted at the beginning of each series of tests, which consists of five test cycles.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing mild steel. The saw blade used shall be recorded. The pendulum system, if any, shall be in the “off” position.
Feed force	The horizontal feed force (force in direction of the cut) applied to the tool shall be (35 ± 5) N. Excessive gripping force shall be avoided. The downward force applied to the tool shall not exceed the force to ensure contact of the base plate with the workpiece by more than 30 N. The feed force and the downward force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 8 mm wide strip across the 100 mm width of the metal sheet. Measurement starts when the saw blade enters the metal sheet and stops when the saw blade leaves the metal sheet.

Table I.103 – Test conditions for sabre saws cutting wooden beams

Orientation	Cutting a horizontal beam of construction wood such as fir with a cross-section of (100 ± 5) mm $\times$ (100 ± 5) mm and a minimum length of 500 mm. The beam shall be supported on resilient material and fixed by screws, clamps, air cylinders or the like to a test rig, see Figure I.102. In all cases, the beam excess end shall be approximately 250 mm from the clamp and shall be readjusted at the beginning of each series of tests, which consists of five test cycles. During the test, the operator may use swivel action to allow proper chip ejection.
Tool bit/settings	New saw blade for the entire series of tests as specified for sawing large wooden beams. The saw blade used shall be recorded. Pendulum systems, if any, being set as specified by the manufacturer for cutting wooden beams. If not specified by the manufacturer, pendulum systems are set to maximum. The guide plate shall be fitted.
Feed force	The vertical feed force (force in direction of the cut) applied to the tool in addition to its weight shall be (40 ± 5) N. Excessive gripping force shall be avoided. The horizontal force applied to the tool shall not exceed the force to ensure contact of the guide plate with the workpiece by more than 30 N. The feed force and the horizontal force shall be determined e.g. by means of a scale and shall be recorded.
Test cycle	Cutting off an approximately 30 mm thick slice. Measurement starts when the saw blade enters the wood and stops when the saw blade leaves the wood.

I.3.6.1 Reported vibration value

Replacement:

Three series of five consecutive tests shall be carried out using a different operator for each series. If it can be shown that the vibration is not affected by operator characteristics, it is acceptable to perform all 15 measurements with one operator only.

The measurements are made in three axes and the results of each direction shall be combined using formula (I.3) to obtain the vibration total value a_{HV} .

The measurement result a_h shall be determined as the arithmetic mean of vibration total values over the tests and operators.

For **jig saws**, the results a_h for two operation modes shall be reported:

- $a_{h,B}$ = mean vibration “cutting board” in accordance with Table I.101.
- $a_{h,M}$ = mean vibration “cutting sheet metal” in accordance Table I.102.

For **sabre saws**, the results a_h for two operation modes shall be reported:

- $a_{h,B}$ = mean vibration “cutting board” in accordance with Table I.101.
- $a_{h,WB}$ = mean vibration “cutting wooden beam” in accordance Table I.103.

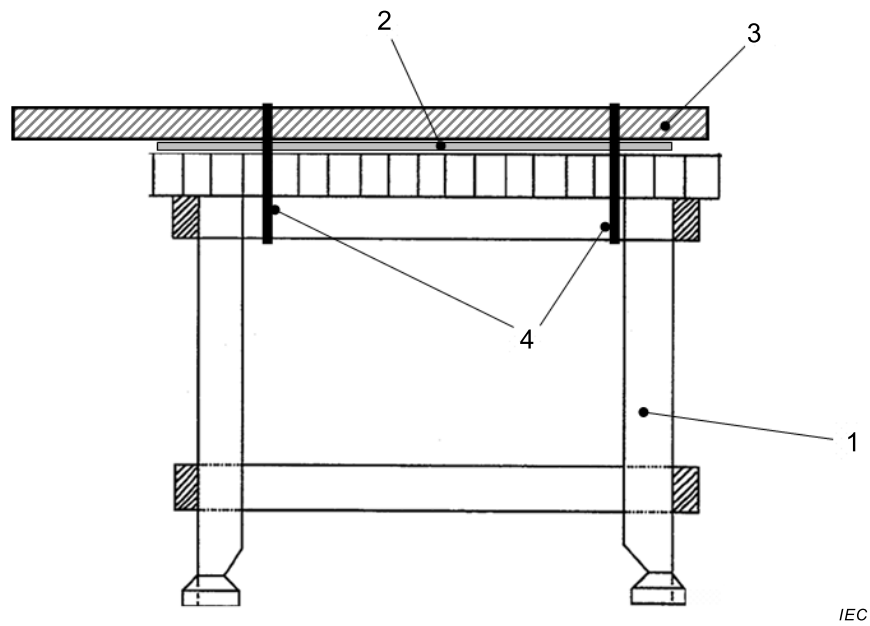
I.3.6.2 Declaration of the vibration total value

Addition:

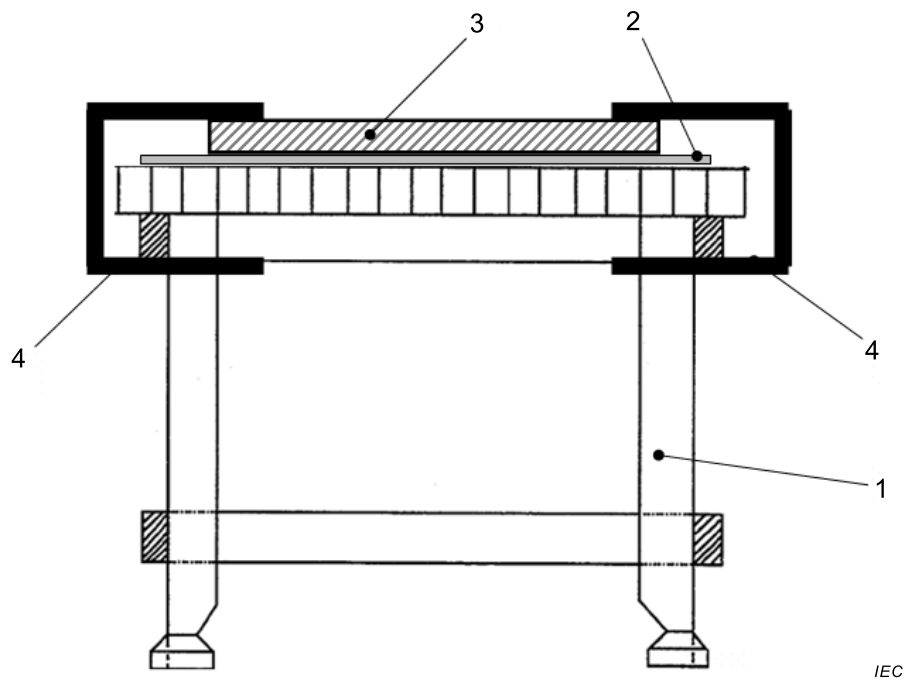
The vibration total value of the handle with the highest emission and the uncertainty K shall be declared together with the saw blade used:

- for **jig saws**
the value of $a_{h,B}$, with the work mode description “cutting boards with saw blade ...” and the value of $a_{h,M}$, with the work mode description “cutting sheet metal with saw blade ...”;
- for **sabre saws**
the value of $a_{h,B}$, with the work mode description “cutting boards with saw blade ...” and the value of $a_{h,WB}$, with the work mode description “cutting wooden beams with saw blade ...”.

NOTE The saw blade significantly influences the vibration behaviour. Therefore the combination of the vibration value along with the saw blade used for the test provides important information.



a) Front view

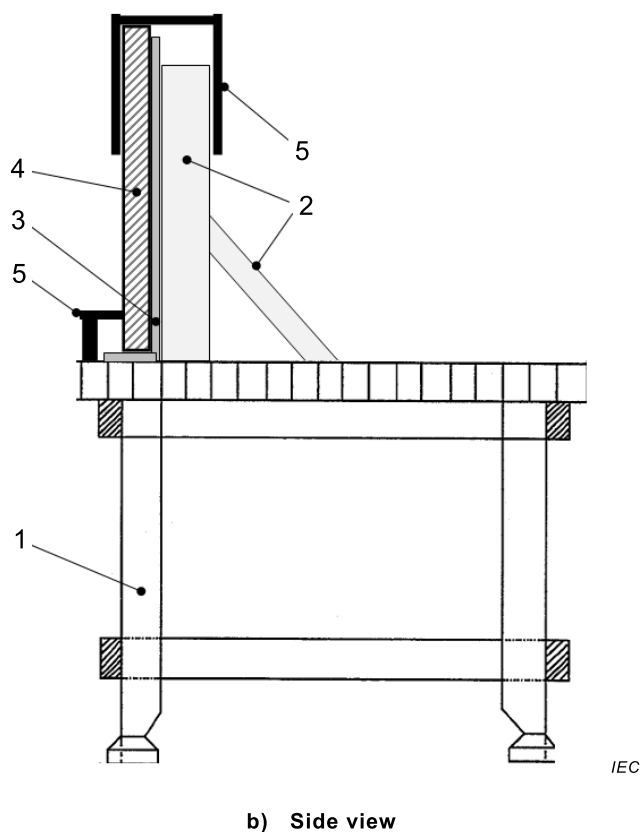
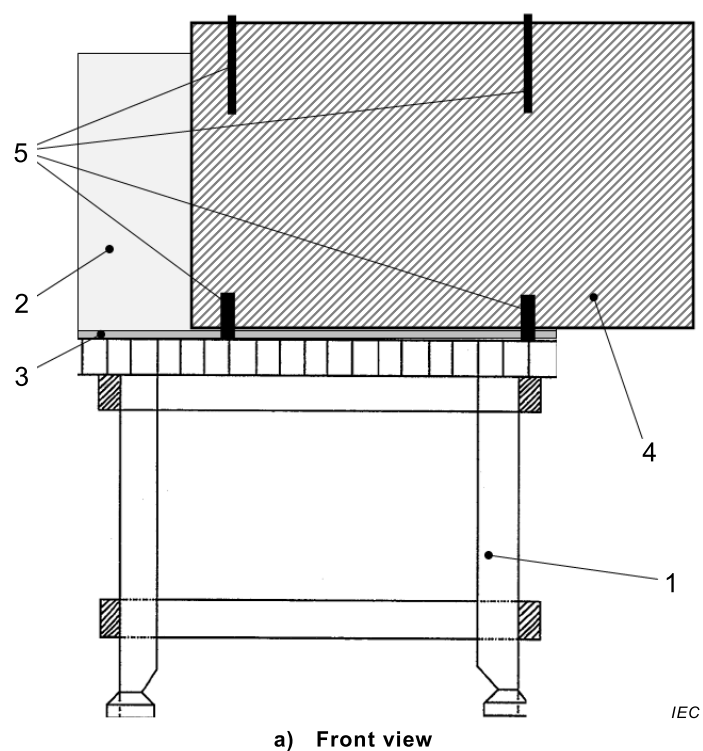


b) Side view

Key

- 1 test bench of Figure I.1
- 2 resilient material
- 3 workpiece
- 4 fixing means (e.g. clamps)

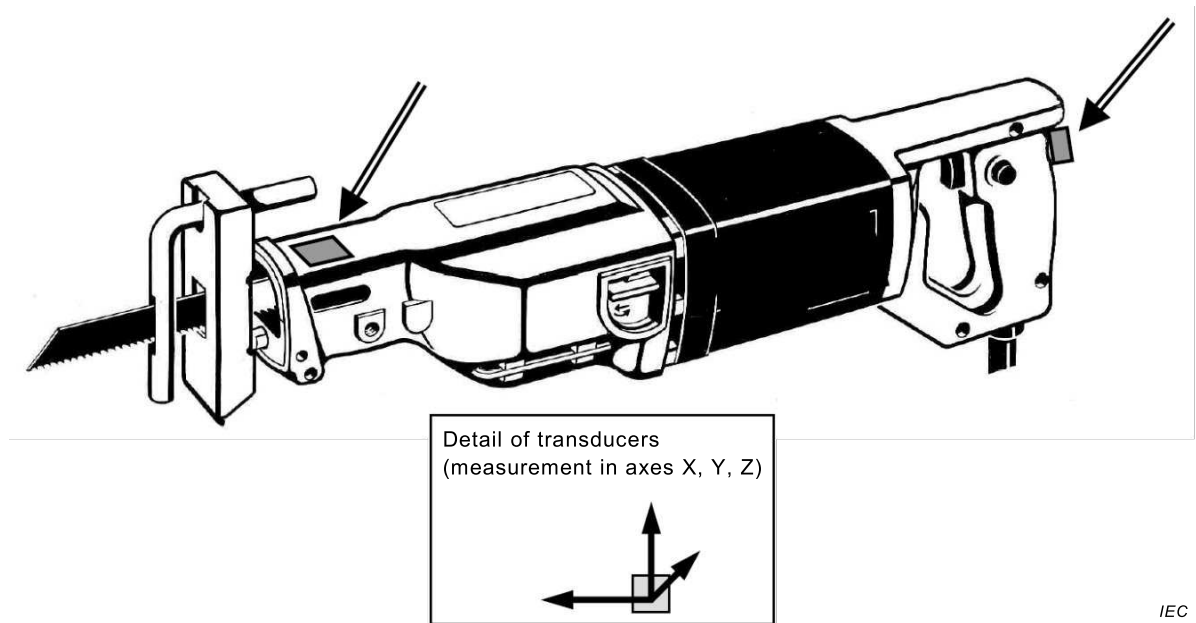
Figure I.101 – Test set-up for jig saws



Key

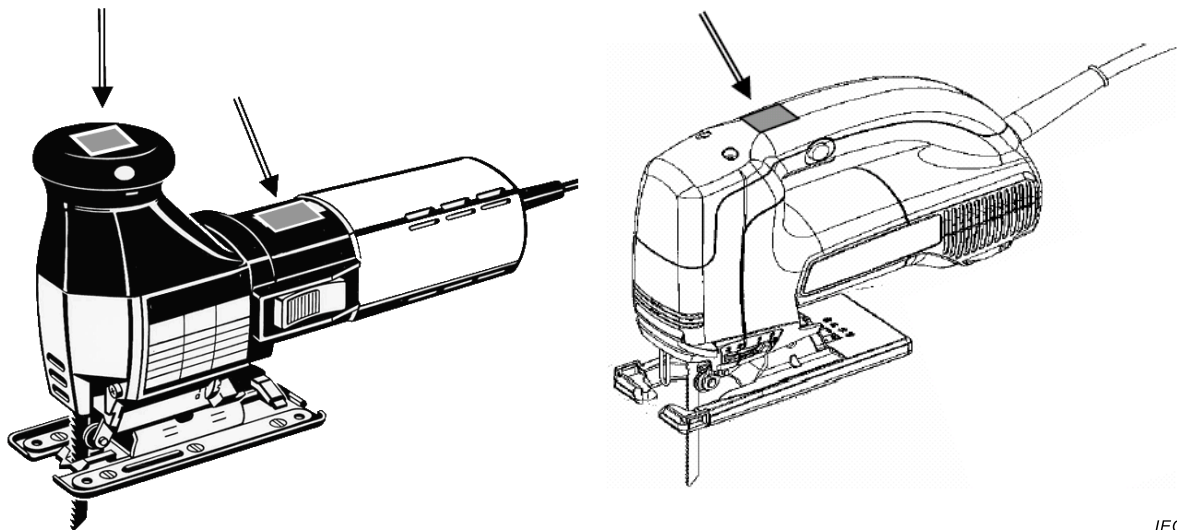
- 1 test bench of Figure I.1
- 2 vertical support, rigidly mounted on test bench
- 3 resilient material
- 4 workpiece
- 5 fixing means (e.g. clamps)

Figure I.102 – Test set-up for sabre saws



IEC

Figure I.103 – Positions of transducers for sabre saws



IEC

Figure I.104 – Positions of transducers for jig saws

Annex K (normative)

Battery tools and battery packs

K.1 Scope

Addition:

All clauses of this Part 2-11 apply unless otherwise specified in this annex.

K.8.14.1.101 *Replacement of item a):*

- a) **Hold the power tool by insulated gripping surfaces, when performing an operation where the cutting accessory may contact hidden wiring. Cutting accessory contacting a "live" wire may make exposed metal parts of the power tool "live" and could give the operator an electric shock.**

NOTE In Europe (EN 62841-2-11), the following additional subclause applies:

K.21.18.Z101 Isolation device

Reciprocating saws with an **integral battery** shall be equipped with an isolation device to prevent the risk of injury from mechanical hazards during **user maintenance** that

- ensures a reliable disconnection of at least one pole of the **battery** from the relevant load(s);
- has a reliable mechanical link between the manual control and the disconnecting element(s);
- is equipped with an unambiguous indication of the state of the disconnection device which corresponds to each position of its manual control (actuator);
- protects against accidental reconnection.

NOTE 101 Examples of methods to achieve this disconnection include removable jumpers, removable **batteries** or a electromechanical **power switch** with a direct mechanical link between the actuator and the contact.

NOTE 102 The risk of accidental reconnection for a **power switch** is addressed by the requirement of 21.18.1.2. The other examples in NOTE 1 achieve this by the necessary actions for reconnection.

Compliance is checked by inspection.

Annex L (normative)

Battery tools and battery packs provided with mains connection or non-isolated sources

L.1 Scope

Addition:

All clauses of this Part 2-11 apply unless otherwise specified in this annex.

NOTE In Europe (EN 62841-2-11), the following additional subclause applies:

L.21.18.Z101 Isolation device

Reciprocating saws with an **integral battery** shall be equipped with a isolation device to prevent the risk of injury from mechanical hazards during **user maintenance** that

- ensures a reliable disconnection of at least one pole of the **battery** from the relevant load(s);
- has a reliable mechanical link between the manual control and the disconnecting element(s);
- is equipped with an unambiguous indication of the state of the disconnection device which corresponds to each position of its manual control (actuator);
- protects against accidental reconnection.

NOTE 101 Examples of methods to achieve this disconnection include removable jumpers, removable **batteries** or a electromechanical **power switch** with a direct mechanical link between the actuator and the contact.

NOTE 102 The risk of accidental reconnection for a **power switch** is addressed by the requirement of 21.18.1.2. The other examples in NOTE 1 achieve this by the necessary actions for reconnection.

Compliance is checked by inspection.

Bibliography

The bibliography of Part 1 is applicable.

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